GEMM Analytical Model - Equations

Auto-generated from PYTHON_FORMULAS and LATEX_SYMBOL_OVERRIDES in gemmanalytics.py.

Output order: execution

Machine Parameters (Pre-Defined)

$f^{(GT)}_{GHz}$: GT Freq (GHz)

$|XeCU|$: XeCU count

$|XeCore|_{perXeCU}$: XeCore per XeCU

$|EU|_{perXeCore}$: EU per XeCore

$|L2Banks|_{perXeCU}$: L2 banks per XeCU

$L2BankSize_{MB}$: bank capacity (MB)

D_{DPAS} : DPAS depth

$\eta_{systolic}$: compute efficiency %

$L1RdBW^{max}_{Bp(Clk \cdot EU)}$: L1 read max (B/EU/clk)

$L1WrBW^{max}_{Bp(Clk \cdot EU)}$: L1 write max (B/EU/clk)

$GtiRdBW^{max}_{BpClk}$: GTI read max BW (B/clk)

$GtiWrBW^{max}_{BpClk}$: GTI write max BW (B/clk)

$MemBW^{max}_{GBps}$: max possible HBM BW (GB/s)

Workload Parameters (Pre-Defined)

$Fmt^{(matA)}$: input A data format (e.g. fp8)

$Fmt^{(matB)}$: input B data format (e.g. fp4)

$Fmt^{(matD\downarrow)}$: output D data format (e.g. fp8)

M_{dim} : M dimension

K_{dim} : K dimension

N_{dim} : N dimension

$Byte^{(matD)}_{perElement}$: output bytes per element (fp32)

ρ_{GT} : machine occupancy %

$M_{perThread}$: M per thread

$K_{perThread}$: K per thread

$N_{perThread}$: N per thread

$W_{\mathrm{thread}}^{(\mathrm{ThreadGroup})}$: TG width in units of thread

$H_{\mathrm{thread}}^{(\mathrm{ThreadGroup})}$: TG height in units of thread

$W_{\mathrm{TG,keep\ cluster\ size\ as\ 4}}^{(\mathrm{XeCoreCluster})}$: XeCore cluster width in units of TG (keep cluster size as 4)

$H_{\mathrm{TG,keep\ cluster\ size\ as\ 4}}^{(\mathrm{XeCoreCluster})}$: XeCore cluster height in units of TG (keep cluster size as 4)

$W_{\mathrm{TG}}^{(\mathrm{XECUTile})}$: XeCU tile width in units of TG

$H_{\mathrm{TG}}^{(\mathrm{XECUTile})}$: XeCU tile height in units of TG

$W_{\mathrm{XeCU}}^{(\mathrm{GPUTile})}$: GPU tile width in XeCU unit

Computed Equations

row 22 - INPUT A BYTES PER ELEMENT

$$\mathrm{Bytes}_{\mathrm{perElement}}^{(\mathrm{matA})} = \mathrm{DataFormatToBytes}[\mathrm{Fmt}^{(\mathrm{matA})}]$$

row 23 - INPUT B BYTES PER ELEMENT

$$\mathrm{Bytes}_{\mathrm{perElement}}^{(\mathrm{matB})} = \mathrm{DataFormatToBytes}[\mathrm{Fmt}^{(\mathrm{matB})}]$$

row 25 - OUTPUT BYTES PER ELEMENT AFTER DOWN CONVERSION

$$\mathrm{Bytes}_{\mathrm{perElement}}^{(\mathrm{matD}\downarrow)} = \mathrm{DataFormatToBytes}[\mathrm{Fmt}^{(\mathrm{matD}\downarrow)}]$$

row 33 - EU COUNT

$$|\mathrm{EU}| = |\mathrm{XeCore}|_{\mathrm{perXeCU}} \times |\mathrm{EU}|_{\mathrm{perXeCore}} \times |\mathrm{XeCU}|$$

row 36a - MMA MAC THROUGHPUT PER EU

$$\tau_{\mathrm{mMACp(Clk\cdot EU)}}^{(\mathrm{peak})} = \frac{4}{max(\mathrm{Bytes}_{\mathrm{perElement}}^{(\mathrm{matA})}, \mathrm{Bytes}_{\mathrm{perElement}}^{(\mathrm{matB})})} \times \mathrm{D}_{\mathrm{DPAS}} \times 16$$

row 36 - MMA MAC THROUGHPUT PER XECORE

$$\tau_{\mathrm{mMACp(Clk\cdot XeCore)}}^{(\mathrm{peak})} = \tau_{\mathrm{mMACp(Clk\cdot EU)}}^{(\mathrm{peak})} \times |\mathrm{EU}|_{\mathrm{perXeCore}}$$

row 42 - CLKS PER DPAS

$$\mathrm{CLKS}_{\mathrm{DPAS}} = \frac{\mathrm{M}_{\mathrm{perThread}} \times \mathrm{K}_{\mathrm{perThread}} \times \mathrm{N}_{\mathrm{perThread}}}{\frac{\tau_{\mathrm{mMACp(Clk\cdot XeCore)}}^{(\mathrm{peak})}}{|\mathrm{EU}|_{\mathrm{perXeCore}}}}$$

row 43 - THREAD WIDTH IN UNITS OF ELEMENTS

$$W_{\mathrm{element}}^{(\mathrm{Thread})} = \mathrm{N}_{\mathrm{perThread}}$$

row 44 - THREAD HEIGHT IN UNITS OF ELEMENTS

$$\mathrm{H}^{(\mathrm{Thread})}_{\mathrm{element}} = \mathrm{M}_{\mathrm{perThread}}$$

row 47 - TG WIDTH IN UNITS OF ELEMENT REALIZED BY MULTIPLE MMA ITERATIONS

$$\mathrm{W}^{(\mathrm{ThreadGroup,realized\ by\ multiple\ MMA\ iterations})}_{\mathrm{element}} = \mathrm{W}^{(\mathrm{ThreadGroup})}_{\mathrm{thread}} \times \mathrm{W}^{(\mathrm{Thread})}_{\mathrm{element}}$$

row 14 - TG TILES IN N

$$|\mathrm{Tiles}|^{(\mathrm{TG})}_{\mathrm{N}} = \frac{\mathrm{N}_{\mathrm{dim}}}{\mathrm{W}^{(\mathrm{ThreadGroup,realized\ by\ multiple\ MMA\ iterations})}_{\mathrm{element}}}$$

row 48 - TG HEIGHT IN UNITS OF ELEMENT

$$\mathrm{H}^{(\mathrm{ThreadGroup})}_{\mathrm{element}} = \mathrm{H}^{(\mathrm{ThreadGroup})}_{\mathrm{thread}} \times \mathrm{H}^{(\mathrm{Thread})}_{\mathrm{element}}$$

row 15 - TG TILES IN M

$$|\mathrm{Tiles}|^{(\mathrm{TG})}_{\mathrm{M}} = \frac{\mathrm{M}_{\mathrm{dim}}}{\mathrm{H}^{(\mathrm{ThreadGroup})}_{\mathrm{element}}}$$

row 52 - XECORE CLUSTER WIDTH IN UNITS OF ELEMENT

$$\mathrm{W}^{(\mathrm{XeCoreCluster})}_{\mathrm{element}} = \mathrm{W}^{(\mathrm{XeCoreCluster})}_{\mathrm{TG,keep\ cluster\ size\ as\ 4}} \times \mathrm{W}^{(\mathrm{ThreadGroup,realized\ by\ multiple\ MMA\ iterations})}_{\mathrm{element}}$$

row 16 - TG CLUSTER TILES IN N

$$|\mathrm{Tiles}|^{(\mathrm{TG\ Cluster})}_{\mathrm{N}} = \frac{\mathrm{N}_{\mathrm{dim}}}{\mathrm{W}^{(\mathrm{XeCoreCluster})}_{\mathrm{element}}}$$

row 53 - XECORE CLUSTER HEIGHT IN UNITS OF ELEMENT

$$\mathrm{H}^{(\mathrm{XeCoreCluster})}_{\mathrm{element}} = \mathrm{H}^{(\mathrm{XeCoreCluster})}_{\mathrm{TG,keep\ cluster\ size\ as\ 4}} \times \mathrm{H}^{(\mathrm{ThreadGroup})}_{\mathrm{element}}$$

row 17 - TG CLUSTER TILES IN M

$$|\mathrm{Tiles}|^{(\mathrm{TG\ Cluster})}_{\mathrm{M}} = \frac{\mathrm{M}_{\mathrm{dim}}}{\mathrm{H}^{(\mathrm{XeCoreCluster})}_{\mathrm{element}}}$$

row 56 - XECU TILE WIDTH IN UNITS OF ELEMENT

$$\mathrm{W}^{(\mathrm{XeCUTile})}_{\mathrm{element}} = \mathrm{W}^{(\mathrm{XECUTile})}_{\mathrm{TG}} \times \mathrm{W}^{(\mathrm{ThreadGroup,realized\ by\ multiple\ MMA\ iterations})}_{\mathrm{element}}$$

row 18 - XECU TILES IN N

$$|\mathrm{Tiles}|^{(\mathrm{XeCU})}_{\mathrm{N}} = \frac{\mathrm{N}_{\mathrm{dim}}}{\mathrm{W}^{(\mathrm{XeCUTile})}_{\mathrm{element}}}$$

row 20 - GPU TILES IN N

$$|\mathrm{Tiles}|_{\mathrm{N}}^{(\mathrm{GPU})} = \frac{\frac{\mathrm{N}_{\mathrm{dim}}}{\mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCUTile})}}}{\mathrm{W}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})}}$$

row 57 - XECU TILE HEIGHT IN UNITS OF ELEMENT

$$\mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})} = \mathrm{H}_{\mathrm{TG}}^{(\mathrm{XECUTile})} \times \mathrm{H}_{\mathrm{element}}^{(\mathrm{ThreadGroup})}$$

row 19 - XECU TILES IN M

$$|\mathrm{Tiles}|_{\mathrm{M}}^{(\mathrm{XeCU})} = \frac{\mathrm{M}_{\mathrm{dim}}}{\mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})}}$$

row 59 - GPU TILE HEIGHT IN XECU UINT

$$\mathrm{H}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})} = \frac{|\mathrm{XeCU}|}{\mathrm{W}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})}}$$

row 21 - GPU TILES IN M

$$|\mathrm{Tiles}|_{\mathrm{M}}^{(\mathrm{GPU})} = \frac{\frac{\mathrm{M}_{\mathrm{dim}}}{\mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})}}}{\mathrm{H}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})}}$$

row 13 - WAVES

$$\mathrm{N}_{\mathrm{waves}} = \mathrm{CEILING}(|\mathrm{Tiles}|_{\mathrm{N}}^{(\mathrm{GPU})}, 1) \times \mathrm{CEILING}(|\mathrm{Tiles}|_{\mathrm{M}}^{(\mathrm{GPU})}, 1)$$

row 60 - GPU TILE WIDTH IN UNITS OF ELEMETNS

$$\mathrm{W}_{\mathrm{element}}^{(\mathrm{GPUTile})} = \mathrm{W}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})} \times \mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCUTile})}$$

row 61 - GPU TILE HEIGHT IN UNITS OF ELEMETNS

$$\mathrm{H}_{\mathrm{element}}^{(\mathrm{GPUTile})} = \mathrm{H}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})} \times \mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})}$$

row 63 - MAT A INPUT SIZE B

$$\mathrm{Size}_{\mathrm{B}}^{(\mathrm{matA})} = \mathrm{M}_{\mathrm{dim}} \times \mathrm{K}_{\mathrm{dim}} \times \mathrm{Bytes}_{\mathrm{perElement}}^{(\mathrm{matA})}$$

row 64 - MAT B INPUT SIZE B

$$\mathrm{Size}_{\mathrm{B}}^{(\mathrm{matB})} = \mathrm{K}_{\mathrm{dim}} \times \mathrm{N}_{\mathrm{dim}} \times \mathrm{Bytes}_{\mathrm{perElement}}^{(\mathrm{matB})}$$

row 65 - MAT C INPUT D OUTPUT SIZE B

$$\mathrm{Size}_{\mathrm{B}}^{(\mathrm{matC,matD})} = \mathrm{M}_{\mathrm{dim}} \times \mathrm{N}_{\mathrm{dim}} \times \mathrm{Bytes}_{\mathrm{perElement}}^{(\mathrm{matD}\downarrow)}$$

row 66 - MAT D INTERMEDIATE SIZE B

$$\mathrm{Size}_{\mathrm{B}}^{(\mathrm{matD}\downarrow)} = \mathrm{M}_{\mathrm{dim}} \times \mathrm{N}_{\mathrm{dim}} \times \mathrm{Byte}_{\mathrm{perElement}}^{(\mathrm{matD})}$$

row 102 - TOTAL L2 SIZE B FOR A SINGLE INSTANCE

$$\mathrm{L2Size}_{\mathrm{B}} = \mathrm{L2BankSize}_{\mathrm{MB}} \times |\mathrm{L2Banks}|_{\mathrm{perXeCU}} \times 1024 \times 1024$$

row 103 - WORKING DATA SET SIZE OF K IN L2 CORRESP 20K CLOCKS OF THREAD DIVERGENCE

$$|\mathrm{WorkingSet}|_{\text{20K clks of thread divergence}}^{(\text{K in L2})} = \min(20000 \times \frac{\mathrm{K}_{\mathrm{perThread}}}{\mathrm{CLKS}_{\mathrm{DPAS}}}, \mathrm{K}_{\mathrm{dim}})$$

row 104 - TOTAL REQUIRED L2 SIZE FOR IDEAL HIT RATE B FOR A SINGLE INSTANCE AND SINGLE WAVE

$$\begin{aligned}\mathrm{TotalRequiredL2Size}_{\text{matB alwayshit}}^{\text{1 instance, 1 wave}} = {} & \mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})} \times \mathrm{Bytes}_{\mathrm{perElement}}^{(\mathrm{matA})} \times |\mathrm{WorkingSet}|_{\text{20K clks of thread divergence}}^{(\text{K in L2})} \\ & + \mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCUTile})} \times \mathrm{Bytes}_{\mathrm{perElement}}^{(\mathrm{matB})} \times |\mathrm{WorkingSet}|_{\text{20K clks of thread divergence}}^{(\text{K in L2})} \\ & + \mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCUTile})} \times \mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})} \times \mathrm{Bytes}_{\mathrm{perElement}}^{(\mathrm{matD}\downarrow)}\end{aligned}$$

row 125 - MAX POSSIBLE HBM BW FREQ B CLK

$$\mathrm{MemBW}_{\mathrm{BpClk}}^{\mathrm{max}} = \frac{\mathrm{MemBW}_{\mathrm{GBps}}^{\mathrm{max}}}{\mathrm{f}_{\mathrm{GHz}}^{(\mathrm{GT})}}$$

row 37a - WORKLOAD MAC PER XECORE

$$\mathrm{WL}_{\mathrm{MAC}}^{(\mathrm{XeCore})} = \frac{\mathrm{M}_{\mathrm{dim}} \times \mathrm{K}_{\mathrm{dim}} \times \mathrm{N}_{\mathrm{dim}}}{|\mathrm{XeCore}|_{\mathrm{perXeCU}} \times |\mathrm{XeCU}|}$$

row 37 - CLK SPECIFIED EFFICIENCY

$$\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})} = \frac{\mathrm{WL}_{\mathrm{MAC}}^{(\mathrm{XeCore})}}{\tau_{\mathrm{mMACp(Clk\cdot XeCore)}}^{(\mathrm{peak})} \times \eta_{\mathrm{systolic}}}$$

row 69 - TOTAL L2 READ B

$$\begin{aligned}\mathrm{L2Rd}_{\mathrm{B}}^{(\mathrm{total})} = {} & \mathrm{Size}_{\mathrm{B}}^{(\mathrm{matA})} \times \mathrm{CEILING}(\frac{\mathrm{N}_{\mathrm{dim}}}{\mathrm{W}_{\mathrm{element}}^{(\text{ThreadGroup,realized by multiple MMA iterations})}}, 1) \\ & + \mathrm{Size}_{\mathrm{B}}^{(\mathrm{matB})} \times \mathrm{CEILING}(\frac{\mathrm{M}_{\mathrm{dim}}}{\mathrm{H}_{\mathrm{element}}^{(\mathrm{ThreadGroup})}}, 1)\end{aligned}$$

row 70 - TOTAL L2 WRITE B

$$\mathrm{L2Wr}_{\mathrm{B}}^{(\mathrm{total})} = \mathrm{Size}_{\mathrm{B}}^{(\mathrm{matC,matD})}$$

row 71 - TOTAL L1 READ B

$$\mathrm{L1Rd}_{\mathrm{B}}^{(\mathrm{total})} = \mathrm{Size}_{\mathrm{B}}^{(\mathrm{matA})} \times \mathrm{CEILING}(\frac{\mathrm{N}_{\mathrm{dim}}}{\mathrm{W}_{\mathrm{element}}^{(\mathrm{Thread})}}, 1)$$
$$+\mathrm{Size}_{\mathrm{B}}^{(\mathrm{matB})} \times \mathrm{CEILING}(\frac{\mathrm{M}_{\mathrm{dim}}}{\mathrm{H}_{\mathrm{element}}^{(\mathrm{Thread})}}, 1)$$

row 72 - TOTAL L1 WRITE B

$$\mathrm{L1Wr}_{\mathrm{B}}^{(\mathrm{total})} = \mathrm{Size}_{\mathrm{B}}^{(\mathrm{matC,matD})}$$

row 74 - L2 READ B XECORE CLK

$$\mathrm{L2RdBW}_{\mathrm{Bp(Clk\cdot XeCore)}} = \frac{\frac{\mathrm{L2Rd}_{\mathrm{B}}^{(\mathrm{total})}}{|\mathrm{XeCU}| \times |\mathrm{XeCore}|_{\mathrm{perXeCU}}}}{\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})}}$$

row 75 - L2 WRITE B XECORE CLK

$$\mathrm{L2WrBW}_{\mathrm{Bp(Clk\cdot XeCore)}} = \frac{\frac{\mathrm{L2Wr}_{\mathrm{B}}^{(\mathrm{total})}}{|\mathrm{XeCU}| \times |\mathrm{XeCore}|_{\mathrm{perXeCU}}}}{\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})}}$$

row 76 - L2 READ WRITE B XECORE CLK

$$\mathrm{L2RdWrBW}_{\mathrm{Bp(Clk\cdot XeCore)}} = \mathrm{L2RdBW}_{\mathrm{Bp(Clk\cdot XeCore)}}$$
$$+\mathrm{L2WrBW}_{\mathrm{Bp(Clk\cdot XeCore)}}$$

row 79 - L1 READ B EU CLK

$$\mathrm{L1RdBW}_{\mathrm{Bp(Clk\cdot EU)}} = \frac{\frac{\mathrm{L1Rd}_{\mathrm{B}}^{(\mathrm{total})}}{|\mathrm{EU}|}}{\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})}}$$

row 77 - L1 READ B XECORE CLK

$$\mathrm{L1RdBW}_{\mathrm{Bp(Clk\cdot XeCore)}} = \mathrm{L1RdBW}_{\mathrm{Bp(Clk\cdot EU)}} \times |\mathrm{EU}|_{\mathrm{perXeCore}}$$

row 80 - L1 WRITE B EU CLK

$$\mathrm{L1WrBW}_{\mathrm{Bp(Clk\cdot EU)}} = \frac{\frac{\mathrm{L1Wr}_{\mathrm{B}}^{(\mathrm{total})}}{|\mathrm{EU}|}}{\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})}}$$

row 78 - L1 WRITE B XECORECLK

$$\mathrm{L1WrBW}_{\mathrm{Bp(Clk\cdot XeCore)}} = \mathrm{L1WrBW}_{\mathrm{Bp(Clk\cdot EU)}} \times |\mathrm{EU}|_{\mathrm{perXeCore}}$$

row 83 - L2 READ MAX B XECORE CLK

$$\mathrm{L2RdBW}^{(\mathrm{max})}_{\mathrm{Bp(Clk \cdot XeCore)}} = \frac{|\mathrm{XeCore}|_{\mathrm{perXeCU}} \times 64}{|\mathrm{XeCore}|_{\mathrm{perXeCU}}}$$

row 84 - L2 WRITE MAX B XECORE CLK

$$\mathrm{L2WrBW}^{(\mathrm{max})}_{\mathrm{Bp(Clk \cdot XeCore)}} = \mathrm{L2RdBW}^{(\mathrm{max})}_{\mathrm{Bp(Clk \cdot XeCore)}}$$

row 85 - L2 READ WRITE MAX B XECORE CLK

$$\mathrm{L2RdWrBW}^{(\mathrm{max})}_{\mathrm{Bp(Clk \cdot XeCore)}} = \mathrm{L2WrBW}^{(\mathrm{max})}_{\mathrm{Bp(Clk \cdot XeCore)}}$$

row 90 - L2 READ B XECORE CLK PCT

$$\eta_{\mathrm{L2RdBWpXeCore}} = \frac{\mathrm{L2RdBW}_{\mathrm{Bp(Clk \cdot XeCore)}}}{\mathrm{L2RdBW}^{(\mathrm{max})}_{\mathrm{Bp(Clk \cdot XeCore)}}}$$

row 91 - L2 WRITE B XECORE CLK PCT

$$\eta_{\mathrm{L2WrBWpXeCore}} = \frac{\mathrm{L2WrBW}_{\mathrm{Bp(Clk \cdot XeCore)}}}{\mathrm{L2WrBW}^{(\mathrm{max})}_{\mathrm{Bp(Clk \cdot XeCore)}}}$$

row 92 - L2 READ WRITE B XECORE CLK PCT

$$\eta_{\mathrm{L2RdWrBWpXeCore}} = \frac{\mathrm{L2RdWrBW}_{\mathrm{Bp(Clk \cdot XeCore)}}}{\mathrm{L2RdWrBW}^{(\mathrm{max})}_{\mathrm{Bp(Clk \cdot XeCore)}}}$$

row 93 - L1 READ B EU CLK PCT

$$\eta_{\mathrm{L1RdBWpEU}} = \frac{\mathrm{L1RdBW}_{\mathrm{Bp(Clk \cdot EU)}}}{\mathrm{L1RdBW}^{\mathrm{max}}_{\mathrm{Bp(Clk \cdot EU)}}}$$

row 94 - L1 WRITE B EU CLK PCT

$$\eta_{\mathrm{L1WrBWpEU}} = \frac{\mathrm{L1WrBW}_{\mathrm{Bp(Clk \cdot EU)}}}{\mathrm{L1WrBW}^{\mathrm{max}}_{\mathrm{Bp(Clk \cdot EU)}}}$$

row 106 - L2 HIT RATE ASSUMED RANDOM ACCESS WITHIN THE WORKING DATA SET PCT

$$\mathrm{L2HitRate}_{\mathrm{random\ WS\ access}} = \min\left(\frac{\mathrm{L2Size_B}}{\mathrm{TotalRequiredL2Size}^{\mathrm{1\ instance,\ 1\ wave}}_{\mathrm{matB\ alwayshit}}}, 1\right)$$

row 107 - L2 MISS RATE PCT

$$\mathrm{L2MissRate} = 1 - \mathrm{L2HitRate}_{\mathrm{random\ WS\ access}}$$

row 108 - TOTAL L2 READ TRAFFIC B

$$\mathrm{L2RdB_{total}} = \mathrm{L2Rd_B^{(total)}}$$

row 110 - PROBABILITY OF MATA HIT IN L2 DURING A NON FIRST WAVE PCT

$$\begin{aligned}\mathrm{P^{(matA)}_{L2Hit,\ after\ 1st\ wave}} = {} & \mathrm{IF(Size_B^{(matA)}} \\ & \mathrm{+Size_B^{(matB)}} \\ & \mathrm{+Size_B^{(matC,matD)}} \le \mathrm{L2Size_B}, 1.0, \mathrm{IF(K_{dim}} > 2 \times |\mathrm{WorkingSet}|^{(\mathrm{K\ in\ L2})}_{\mathrm{20K\ clks\ of\ thread\ divergence}}, 0.0, 1 \\ & -\frac{\mathrm{K_{dim}} - |\mathrm{WorkingSet}|^{(\mathrm{K\ in\ L2})}_{\mathrm{20K\ clks\ of\ thread\ divergence}}}{|\mathrm{WorkingSet}|^{(\mathrm{K\ in\ L2})}_{\mathrm{20K\ clks\ of\ thread\ divergence}}}))\end{aligned}$$

row 111 - PROBABILITY OF MATB HIT IN L2 DURING A NON FIRST WAVE PCT

$$\begin{aligned}\mathrm{P^{(matB)}_{L2Hit,\ after\ 1st\ wave}} = {} & \mathrm{IF(Size_B^{(matA)}} \\ & \mathrm{+Size_B^{(matB)}} \\ & \mathrm{+Size_B^{(matC,matD)}} \le \mathrm{L2Size_B}, 1.0, 0.0)\end{aligned}$$

row 112 - PROBABILITY OF MATA MISS IN L2 DURING A NON FIRST WAVE PCT

$$\begin{aligned}\mathrm{P^{(matA)}_{L2Miss,\ after\ 1st\ wave}} = {} & 1 \\ & -\mathrm{P^{(matA)}_{L2Hit,\ after\ 1st\ wave}}\end{aligned}$$

row 113 - PROBABILITY OF MATB MISS IN L2 DURING A NON FIRST WAVE PCT

$$\begin{aligned}\mathrm{P^{(matB)}_{L2Miss,\ after\ 1st\ wave}} = {} & 1 \\ & -\mathrm{P^{(matB)}_{L2Hit,\ after\ 1st\ wave}}\end{aligned}$$

row 116 - TOTAL HBM READ B AFTER A COMPLETION OF A WAVE CONSIDER COLD CACHE

$$\mathrm{MemRdB^{(wave)}_{cold\ start}} = \frac{\left(\mathrm{MemRdB^{(wave)}_{cold\ start}}\right)_{\mathrm{num}}}{2}$$

$$\left(\mathrm{MemRdB}^{(\mathrm{wave})}_{\text{cold start}}\right)_{\mathrm{num}} = \mathrm{Size}^{(\mathrm{matA})}_{\mathrm{B}}$$
$$+\mathrm{Size}^{(\mathrm{matA})}_{\mathrm{B}} \times \left(\mathrm{CEILING}(\frac{\mathrm{N}_{\mathrm{dim}}}{\mathrm{W}^{(\mathrm{XeCUTile})}_{\mathrm{element}}}, 1) - 1\right) \times \mathrm{P}^{(\mathrm{matA})}_{\text{L2Miss, after 1st wave}}$$
$$+\mathrm{Size}^{(\mathrm{matB})}_{\mathrm{B}}$$
$$+\mathrm{Size}^{(\mathrm{matB})}_{\mathrm{B}} \times \left(\mathrm{CEILING}(\frac{\mathrm{M}_{\mathrm{dim}}}{\mathrm{H}^{(\mathrm{XeCUTile})}_{\mathrm{element}}}, 1) - 1\right) \times \mathrm{P}^{(\mathrm{matB})}_{\text{L2Miss, after 1st wave}}$$
$$+\left(\left(\mathrm{MemRdB}^{(\mathrm{wave})}_{\text{cold start}}\right)_{\mathrm{num}}\right)_{\mathrm{aux4}} \times \mathrm{L2MissRate}$$

$$\left(\left(\mathrm{MemRdB}^{(\mathrm{wave})}_{\text{cold start}}\right)_{\mathrm{num}}\right)_{\mathrm{aux4}} = \mathrm{L2RdB}_{\mathrm{total}}$$
$$-\left(\left(\left(\mathrm{MemRdB}^{(\mathrm{wave})}_{\text{cold start}}\right)_{\mathrm{num}}\right)_{\mathrm{aux4}}\right)_{\mathrm{aux1}}$$

$$\left(\left(\left(\mathrm{MemRdB}^{(\mathrm{wave})}_{\text{cold start}}\right)_{\mathrm{num}}\right)_{\mathrm{aux4}}\right)_{\mathrm{aux1}} = \mathrm{Size}^{(\mathrm{matA})}_{\mathrm{B}}$$
$$+\mathrm{Size}^{(\mathrm{matA})}_{\mathrm{B}} \times \left(\mathrm{CEILING}(\frac{\mathrm{N}_{\mathrm{dim}}}{\mathrm{W}^{(\mathrm{XeCUTile})}_{\mathrm{element}}}, 1) - 1\right) \times \mathrm{P}^{(\mathrm{matA})}_{\text{L2Miss, after 1st wave}}$$
$$+\mathrm{Size}^{(\mathrm{matB})}_{\mathrm{B}}$$
$$+\mathrm{Size}^{(\mathrm{matB})}_{\mathrm{B}} \times \left(\mathrm{CEILING}(\frac{\mathrm{M}_{\mathrm{dim}}}{\mathrm{H}^{(\mathrm{XeCUTile})}_{\mathrm{element}}}, 1) - 1\right) \times \mathrm{P}^{(\mathrm{matB})}_{\text{L2Miss, after 1st wave}}$$

row 117 - TOTAL HBM WRITE B

$$\mathrm{MemWrB}^{(\mathrm{total})} = \mathrm{Size}^{(\mathrm{matC,matD})}_{\mathrm{B}}$$

row 118 - TOTAL HBM B

$$\mathrm{MemRdWrB}^{(\mathrm{total})} = \mathrm{MemWrB}^{(\mathrm{total})} + \mathrm{MemRdB}^{(\mathrm{wave})}_{\text{cold start}}$$

row 121 - HBM READ B CLK

$$\mathrm{MemRdBW}_{\mathrm{BpClk}} = \frac{\mathrm{MemRdB}^{(\mathrm{wave})}_{\text{cold start}}}{\mathrm{T}^{(\mathrm{total})}_{\mathrm{clk}}}$$

row 122 - HBM WRITE B CLK

$$\mathrm{MemWrBW}_{\mathrm{BpClk}} = \frac{\mathrm{MemWrB}^{(\mathrm{total})}}{\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})}}$$

row 123 - HBM TOTAL B CLK

$$\mathrm{MemRdWrBW}_{\mathrm{BpClk}} = \frac{\mathrm{MemRdWrB}^{(\mathrm{total})}}{\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})}}$$

row 126 - HBM BW PCT

$$\eta_{\mathrm{MemBW}} = \frac{\mathrm{MemRdWrBW}_{\mathrm{BpClk}}}{\mathrm{MemBW}_{\mathrm{BpClk}}^{\mathrm{max}}}$$

row 127 - GTI READ BW PCT

$$\eta_{\mathrm{GTIRdBW}} = \frac{\mathrm{MemRdBW}_{\mathrm{BpClk}}}{\mathrm{GtiRdBW}_{\mathrm{BpClk}}^{\mathrm{max}}}$$

row 128 - GTI WRITE BW PCT

$$\eta_{\mathrm{GTIWrBW}} = \frac{\mathrm{MemWrBW}_{\mathrm{BpClk}}}{\mathrm{GtiWrBW}_{\mathrm{BpClk}}^{\mathrm{max}}}$$